Glossary

Purpose

This glossary defines the canonical meaning of architectural terms used throughout the Shio Runtime Architecture Specification.

Every term has exactly one architectural meaning.

Whenever implementation details differ from these definitions, the glossary takes precedence.

Runtime

An independent subsystem responsible for one well-defined aspect of the digital character.

A Runtime owns:

- one responsibility;
- its lifecycle;
- its resources;
- its internal implementation.

A Runtime communicates through Character State, Event Bus and public contracts.

Responsibility

A clearly defined architectural concern owned by exactly one Runtime.

Responsibilities must never overlap.

Every feature should naturally belong to one Runtime.

Character

The digital entity experienced by the user.

Shio is the character.

The Runtime exists to coordinate the systems that make the character believable.

Character State

The single source of truth describing the current observable condition of the character.

Character State represents what **is true now**.

It is persistent.

It is observable.

It is never used to describe historical events.

Event

An immutable notification describing something that has already happened.

Events represent history.

Character State represents the present.

Event Bus

The communication mechanism responsible for delivering Events between Runtime systems.

The Event Bus does not store state.

It distributes completed facts.

Runtime Boundary

The point where one Runtime's responsibility ends and another Runtime's responsibility begins.

Responsibilities cross boundaries.

Implementation does not.

Runtime Contract

The architectural agreement that every Runtime follows.

It defines ownership, lifecycle, communication rules and responsibilities.

Adapter

A translation layer between the Runtime and an external technology.

Adapters translate.

They never make architectural decisions.

External World

Everything outside Shio Runtime.

Examples include:

- operating systems;
- APIs;
- streaming platforms;
- AI providers;
- databases;
- local services.

Only Action Runtime communicates with the External World.

Action

A request to produce an observable side effect outside the Runtime.

Actions are validated, authorized and executed by Action Runtime.

Tool

A concrete executable capability exposed by Action Runtime.

Examples include:

- BanViewer;
- PlaySound;
- ChangeScene;
- ToggleOverlay.

Tools are implementation mechanisms.

Actions are architectural concepts.

Action Runtime

The only Runtime permitted to modify the External World.

Every external side effect must pass through Action Runtime.

Experience

The emotional interpretation of completed events.

Experience occurs after execution.

It determines how Shio reacts rather than what Shio does.

Experience Layer

The Runtime responsible for transforming completed events into believable character behavior.

Execution changes the world.

Experience changes the character.

Personality

The persistent identity of Shio.

Personality defines who the character is.

It is independent from the language model.

Emotion

The current emotional state of the character.

Emotion changes frequently.

Personality changes slowly.

Memory

The accumulated experiences that preserve continuity across interactions.

Memory belongs to the character.

Not to the language model.

Memory Layer

The Runtime responsible for preserving continuity over time.

Memory Layer determines:

- what should be remembered;
- when it should be remembered;
- when it should be forgotten.

Speech Runtime

The Runtime responsible for verbal communication between the character and the outside world.

It owns the complete communication pipeline from input speech to synthesized output.

Avatar Runtime

The Runtime responsible for the physical embodiment of the character.

It transforms Character State into visible behavior.

Decision Pipeline

The sequence of architectural layers responsible for determining how incoming input should be processed.

It progressively reduces uncertainty before reaching the Character LLM.

Rule Gate

A deterministic decision layer responsible for resolving simple situations without involving a language model.

Rule Gate prefers rules over reasoning whenever possible.

Intent Router

The lightweight classifier responsible for routing incoming communication.

It performs routing.

It does not perform reasoning.

Character LLM

The primary language model responsible for expressing the personality of Shio.

The Character LLM proposes intentions.

It never directly executes actions.

Shared Character State

The common observable state used by every Runtime to coordinate behavior.

It replaces direct Runtime-to-Runtime dependencies.

External Adapter

An Adapter responsible for communicating with one specific external technology or platform.

Examples include:

- LLM Adapters;
- Speech Adapters;
- Platform Adapters;
- Storage Adapters.

Illusion of Life

The primary architectural objective of Shio Runtime.

Every Runtime contributes to making the character appear alive, emotionally consistent and believable.

Whenever architectural decisions are made, preserving this illusion should take priority over exposing implementation details.

Believability

The degree to which the user perceives Shio as a coherent digital character rather than a collection of software components.

Believability is considered a primary architectural quality attribute.

Composition

The practice of creating complex behavior through cooperation between independent Runtime systems instead of introducing monolithic controllers.

Composition is preferred over complexity.

Technology

Any implementation detail used to realize the architecture.

Examples include:

- Groq;
- OpenAI;
- Electron;
- Three.js;
- SQLite;
- Python Speech.

Technologies implement responsibilities.

They never define them.

Architectural Principle

A rule that defines how the Runtime should be designed and evolved.

Architectural Principles are considered stable and should outlive individual technologies.

Canonical Rule

Whenever uncertainty exists about the meaning of a term used within this specification, the definitions contained in this glossary should be considered authoritative.

The glossary establishes a shared architectural language for both developers and AI-assisted implementation tools.